

EXP36 Recognize watch from the given image by general Object recognition using OpenCV.

AIM:- Recognize watch from the given image by general Object recognition

PROGRAM:

```
import cv2

import numpy as np

img = cv2.imread("watch.PNG")

if img is None:

    print("Image not found!")

    print("Make sure the file name is watch.PNG")

    exit()

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

gray = cv2.GaussianBlur(gray, (9, 9), 2)

circles = cv2.HoughCircles(

    gray,

    cv2.HOUGH_GRADIENT,

    dp=1,

    minDist=50,

    param1=100,

    param2=35,

    minRadius=20,

    maxRadius=100

)

if circles is not None:

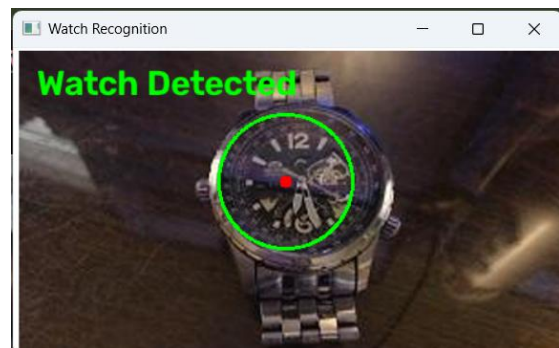
    circles = np.uint16(np.around(circles))
```

```
x, y, r = circles[0][0]
cv2.circle(img, (x, y), r, (0, 255, 0), 2)
cv2.circle(img, (x, y), 5, (0, 0, 255), -1)
cv2.putText(img, "Watch Detected", (20, 40),
            cv2.FONT_HERSHEY_SIMPLEX, 1,
            (0, 255, 0), 2)
else:
    print("No watch circle detected")
cv2.imshow("Watch Recognition", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

INPUT:



OUTPUT:



RESULT:- Thus the program is executed successfully.